LYRA FLETCHER

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MECHATRONICS & ROBOTICS HARDWARE | SUMMER 2027 CO-OP

EDUCATION

University of British Columbia — B.A.Sc., Mechanical Engineering (Mechatronics) **Expected May 2028**

Minor in Entrepreneurship | Cumulative average: 87.7% | Schulich Leader Scholarship ($80,000) | McEwen Family Entrance Award ($25,000)

TECHNICAL SKILLS

CAD & Simulation: SolidWorks, Creo, ANSYS Mechanical/Fluent, MATLAB, Simulink, TracePro; FEA/CFD

Programming & Embedded: Python, C/C++, ESP32, Raspberry Pi, ArduPilot, Pixhawk 6C; data acquisition

Electronics & Controls: Altium (basic PCB design), circuit prototyping, sensor/servo integration, VHDL, PLCs, ladder logic

Fabrication & Testing: CNC, TIG/MIG welding, 3D printing, DFMA, structural/thermal testing

ENGINEERING EXPERIENCE

Airmetal — Electromechanical Engineering Co-op / Acting Mechanical Team Lead **May–Aug 2026**

Ottawa, ON | Rapid-prototyping UAV startup

- Developed ESP32 instrumentation for bending loads, vibration and acceleration; processed sensor data in Python to guide structural design.
- Integrated commercial sensors, Pixhawk 6C, servo actuation and telemetry into 4 UAV prototypes; acquired and analyzed flight data.
- Built a linear-actuator test rig; evaluated tear propagation and puncture resistance in batches of up to 20 samples at varied temperatures.
- Compared prototype bending loads with theoretical predictions to refine lightweight construction, reinforcement and weight distribution.
- Served as Acting Mechanical Team Lead, coordinating a 4-person team, technical reviews and engineering priorities.

NETGEAR R&D — Mechanical Engineering Co-op **May–Dec 2025**

Richmond, BC

- Calibrated ANSYS thermal models against measured hardware temperatures, achieving agreement with experimental results.
- Tested 2 products under torsion and three-point bending; developed ANSYS structural models using measured material properties.
- Designed FEA-validated PCB connector tooling (15% faster installation) and production fixtures (30% faster assembly).
- Designed, optically simulated and released injection-moulded light pipes for 8 commercial products.

ENGINEERING LEADERSHIP

UBC Thunderbikes — Mechanical Team Lead **Sep 2023–Present**

Mechanical Team Lead (2025–Present); Cooling & Thermal Management Lead (2024–2025)

- Lead a 30-person mechanical team for MotoStudent Spain 2027, with weekly technical meetings and 2 design reviews per term.
- Led a 4-person team developing a 16 kW liquid-cooling system (+41% YoY), including 8 kW battery cooling plates.
- Model battery and cooling-plate thermal performance in Python, MATLAB and ANSYS; compare predictions with physical tests.

SELECTED PROJECT

Automated Naloxone Injector — Independent Biomedical Mechatronics Project **2020–2023**

- Prototyped a wearable naloxone-delivery device integrating pulse-oximetry sensing, Arduino control and PWM actuator deployment.